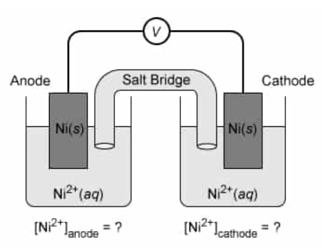
A concentration cell is constructed using Ni(*s*) strips as the anode and the cathode with each immersed in a Ni^{2+}(*aq*) solution of a different concentration, as shown below.

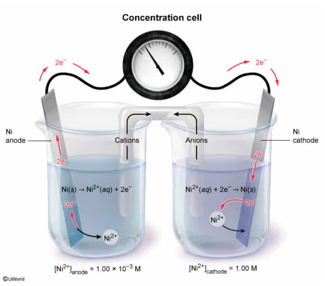


Which of the following statements correctly compares the molar concentrations $[Ni^{2+}]_{anode}$ and $[Ni^{2+}]_{cathode}$ of each half-cell solution and gives the correct direction of electron flow within the cell?

- A. $[Ni^{2+}]_{anode} < [Ni^{2+}]_{cathode}$; electrons flow from cathode to anode [9%]
- B. $[Ni^{2+}]_{cathode} < [Ni^{2+}]_{anode}$; electrons flow from anode to cathode [43%]
- ✔ C. $[Ni^{2+}]_{anode} < [Ni^{2+}]_{cathode}$; electrons flow from anode to cathode [40%]
- D. $[Ni^{2+}]_{cathode} < [Ni^{2+}]_{anode}$; electrons flow from cathode to anode [6%]

Correct
39% answered correctly

Explanation:



In any electrochemical cell, electrons always flow from the anode (ie, the electrode where oxidation occurs) to the cathode (ie, the electrode where reduction occurs). In a **concentration cell**, the **anode** and the **cathode** are both made of the **same metal** and are surrounded by the **same ionic solution**, but the solutions **differ in concentration**. As a result, the same half-reaction occurs in opposite directions at both electrodes.

The concentration cell described in this question uses Ni(*s*) strips as both the anode and the cathode. At the anode, Ni(*s*) atoms are oxidized into Ni^{2+}(*aq*) ions and liberate 2 electrons in each reaction.

$$\text{Anode (oxidation):}\ \ \mathbf{Ni}(s) \rightarrow \mathbf{Ni^{2+}}(aq) + \mathbf{2\ e^-}$$

The liberated electrons migrate to the cathode where Ni^{2+}(*aq*) ions in solution take the electrons and are reduced back to Ni(*s*).

$$\text{Cathode (reduction):}\ \ \mathbf{Ni^{2+}}(aq) + \mathbf{2\ e^-} \rightarrow \mathbf{Ni}(s)$$

Accordingly, concentration cells produce a flow of electrons caused by the drive to equalize the ion concentrations in the two half-cells. In the less-concentrated half-cell, cations are made (ie, by oxidizing metal atoms), and in the more-concentrated half-cell, cations are consumed (ie, by reduction) until the cation concentrations in each half-cell are equal. As such, electrons flow from anode to cathode until both cells have the same Ni^{2+}(*aq*) ion concentration, at which point the reaction stops.

Therefore, the anode must be immersed in the less-concentrated solution (where Ni^{2+}(*aq*) can be produced), and the cathode must be immersed in the more-concentrated solution (where Ni^{2+}(*aq*) can be consumed).

$$\mathbf{[Ni^{2+}]_{anode} < [Ni^{2+}]_{cathode}}$$

For example, this properly configured electrode arrangement with $[Ni^{2+}]_{anode} = 1.00 \times 10^{-3}$ M and $[Ni^{2+}]_{cathode} = 1.00$ M causes electrons to flow from the **anode to the cathode**.

(Choices A and D) In any electrochemical cell, electrons always flow from the anode (the electrode of oxidation) to the cathode (the electrode of reduction).

(Choice B) If $[Ni^{2+}]_{anode}$ were greater than $[Ni^{2+}]_{cathode}$, the cell would not operate because the concentrations in the cells could not progress toward equality.

Educational objective:
In a concentration cell, the anode and the cathode are both made of the same metal and are surrounded by the same ionic solution but the solutions differ in concentration, which produces a flow of electrons caused by the drive to equalize the ion concentrations in the two half-cells. Electrons migrate from the anode (placed in a solution with fewer cations) to the cathode (placed in a solution with more cations) until the concentrations are equal.

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